Remote Procedure Call (RPC)

Outline

- Introduction
- Protocol Stack
- Presentation Formatting

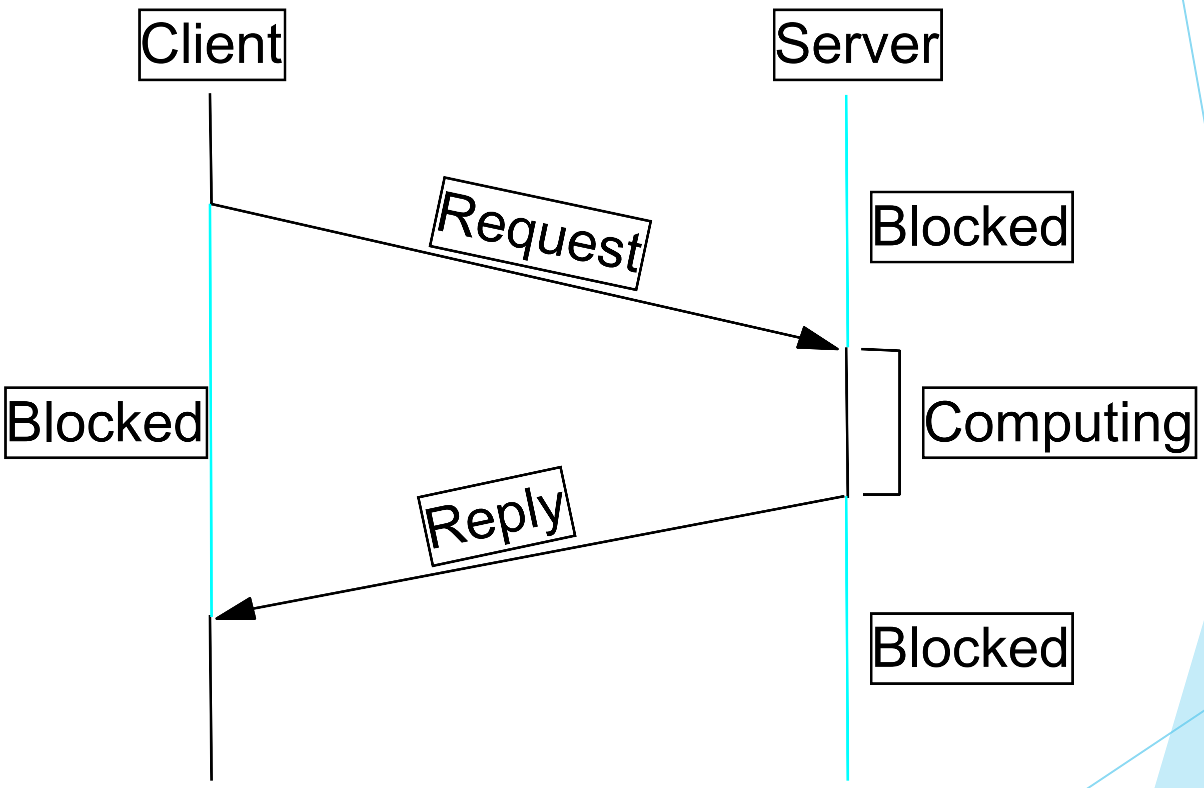
Introduction

- **RPC:** Call a procedure (function, subroutine, method, …) in a program
  - running on a remote machine,
  - while hiding communication details from the programmer.

- Why are TCP or UDP protocols not sufficient for RPC transactions?

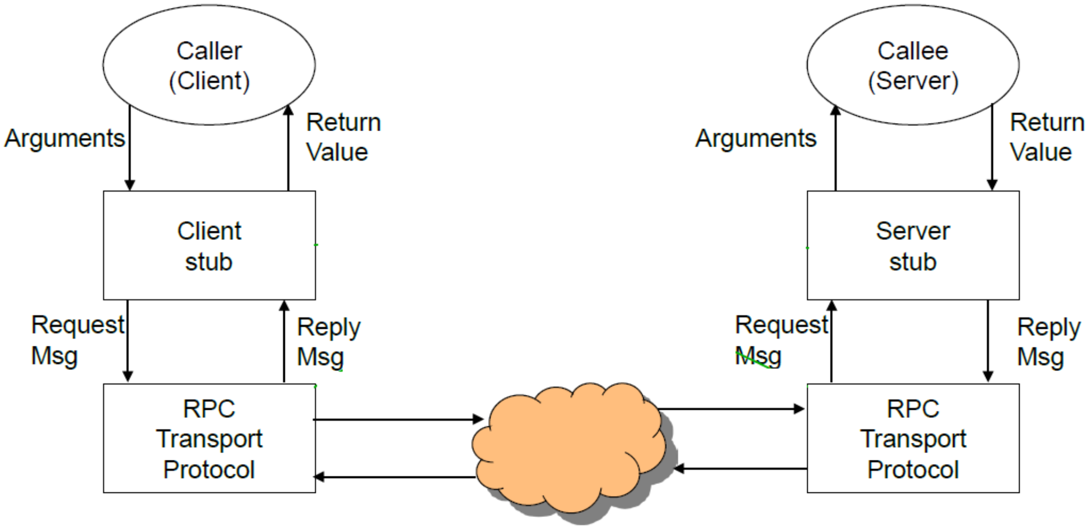
- RFC 1057: Remote Procedure Call
- RFC 1014: External Data Representation

RPC Timeline



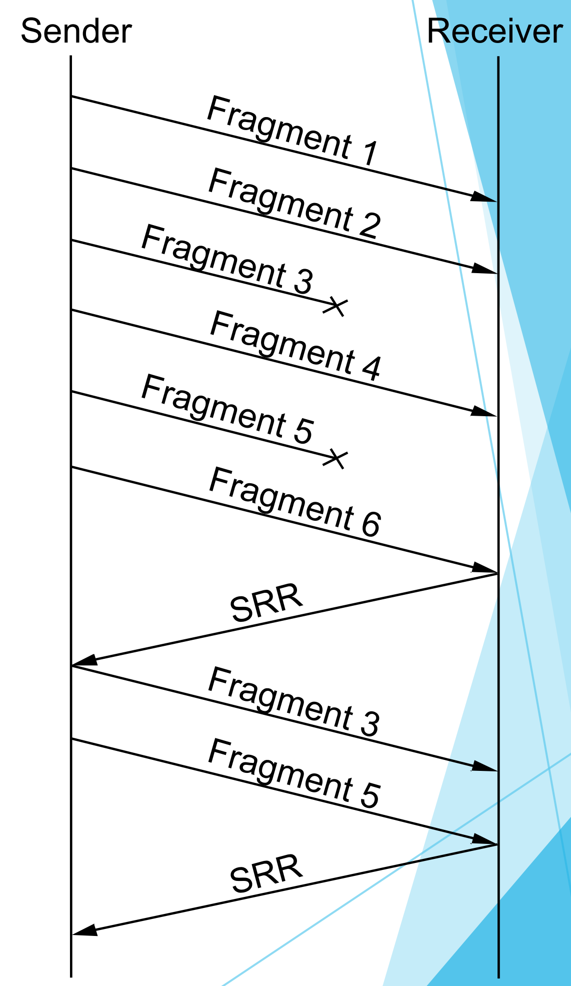
RPC Components

- Stubs
- Protocol Stack
  - BLAST: fragment:
  - CHAN: synchroni:
  - SELECT: dispatch



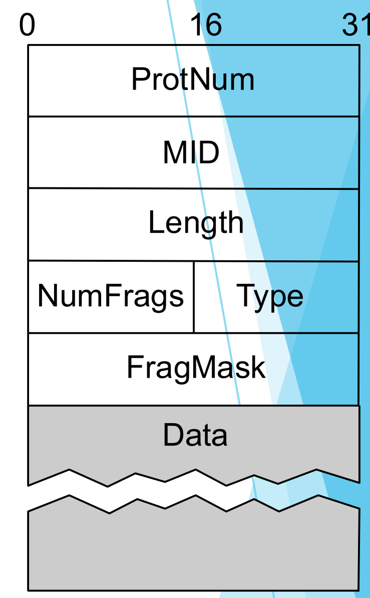
Bulk Transfer (BLAST)

- Unlike IP, tries to recover from lost fragments
  - still no guarantee of message delivery
- Strategy
  - selective retransmission (by SRR)
  - aka partial acknowledgements



BLAST Header Format

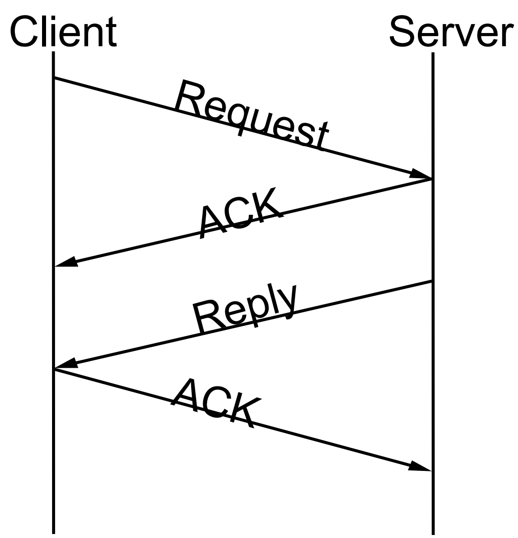
- MID must protect against wrap around
- TYPE = DATA or SRR
- NumFrags indicates number of fragments
- FragMask distinguishes among fragments
  - if Type=DATA, identifies this fragment
  - if Type=SRR, identifies missing fragments



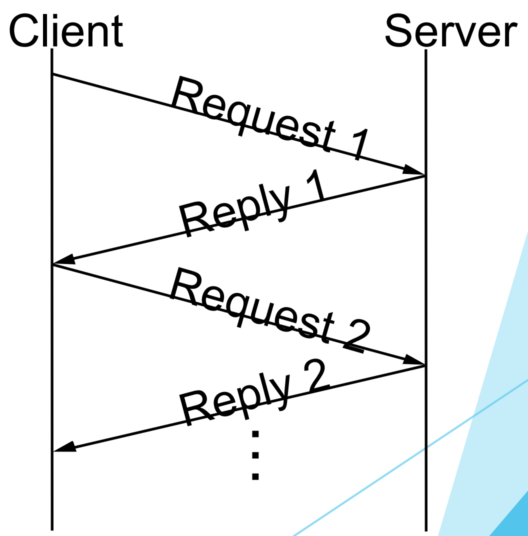
Request/Reply (CHAN)

- Guarantees message delivery
- Synchronizes client with server

Simple case

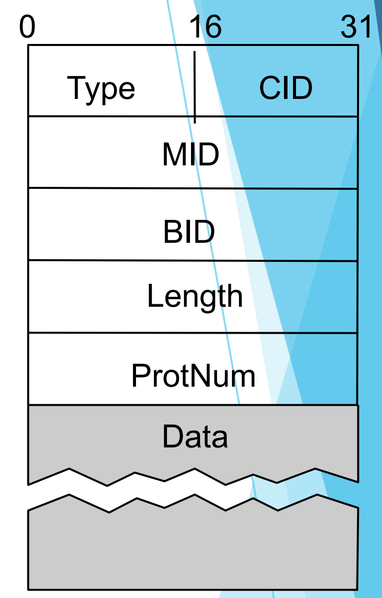


Implicit Acks



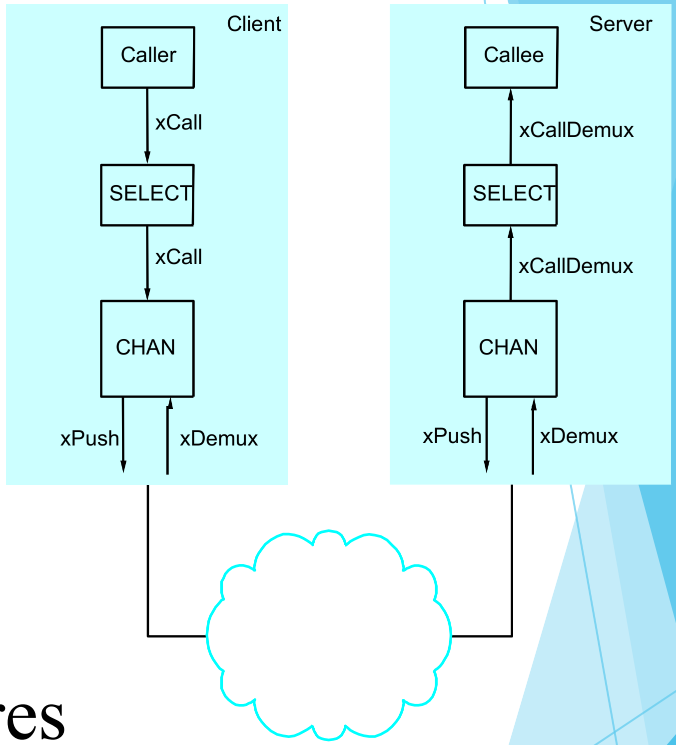
CHAN Header Format

- Type = REQ, REP, ACK, PROBE
- CID = Channel ID, 64K concurrent logical channels between any pair of hosts
- MID identifies each request/reply pair
- BID = Boot ID, incremented each time the machine reboots



Dispatcher (SELECT)

- Dispatch to appropriate procedure
- Synchronous counterpart to UDP
- Managing Concurrency

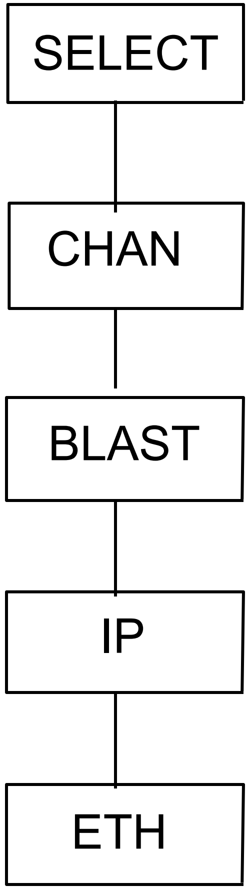


- Address Space for Procedures
  - flat: unique id for each possible procedure
  - hierarchical: program + procedure number

Examples of Hierarchical Procedure Numbers

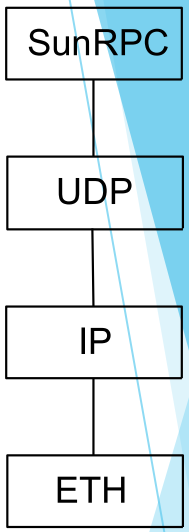
- Program: file server, name server
- Within file server:
  - 1: read, 2: write, 3: seek
- Within name server:
  - 1: insert, 2: lookup

Simple RPC Stack



SunRPC

- Sun’s Network File System (NFS)
- IP implements BLAST-equivalent
  - except no selective retransmit
- SunRPC implements CHAN-equivalent
  - except not at-most-once
- UDP + SunRPC implement SELECT-equivalent
  - UDP dispatches to program (ports bound to programs)
  - SunRPC dispatches to procedure within program



SunRPC

- NFS Program number 0x00100003
  - Getattr: 1
  - Setattr: 2
  - Read: 6
  - Write: 8
- Port Mapper: maps program numbers to port numbers
  - has a program number 0x00100000.
  - run at a well-know port number 111

Presentation Formatting

- Marshalling (encoding) application data into messages
- Unmarshalling (decoding) messages into application data

- Data types we consider
  - integers
  - floats
  - strings
  - arrays
  - structs

- Types of data we do not consider
  - images
  - video
  - multimedia documents

Difficulties

- Representation of base types
  - floating point: IEEE 754 versus non-standard
  - integer: big-endian versus little-endian (e.g., 34,677,374)